



CS & IT ENGINEERING

Discrete Mathematics

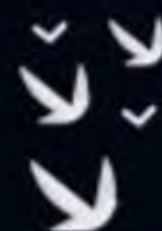
Graph Theory

Lecture No.- 13



By- Satish Sir

Recap of Previous Lecture



Topic

Coloring in Graphs



Topics to be Covered

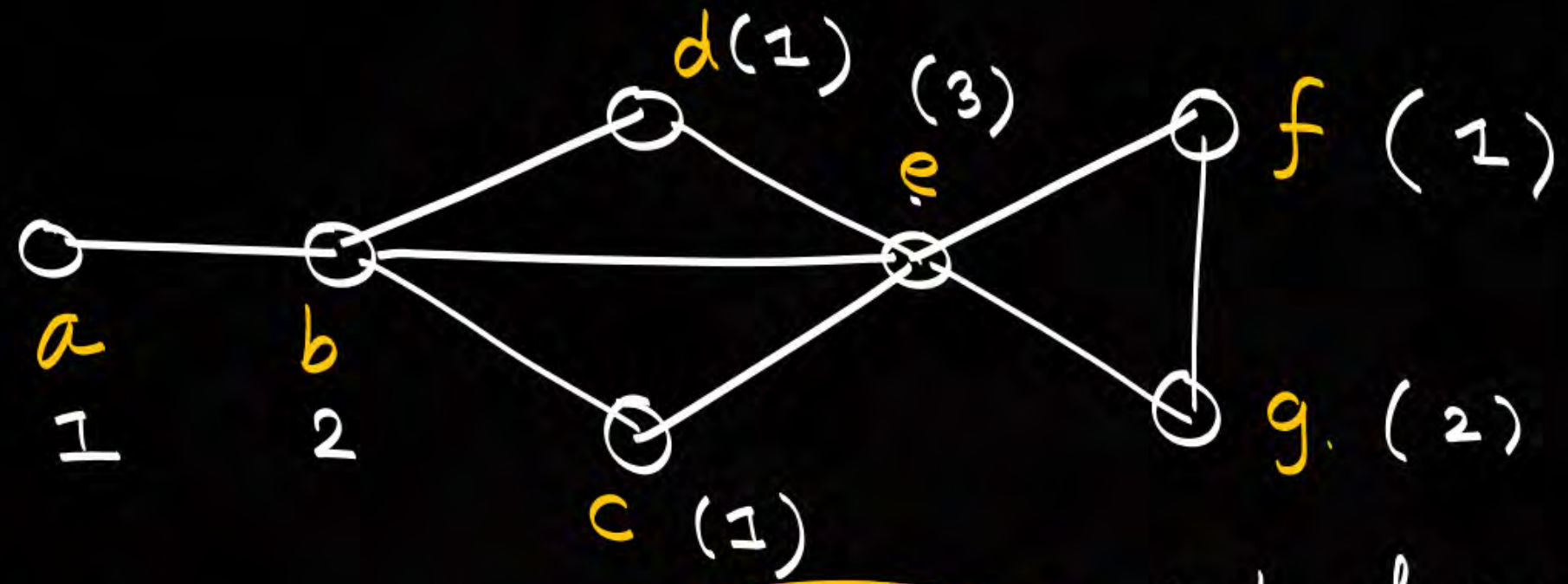


Topic

Dominating Set

Topic

Matching Set

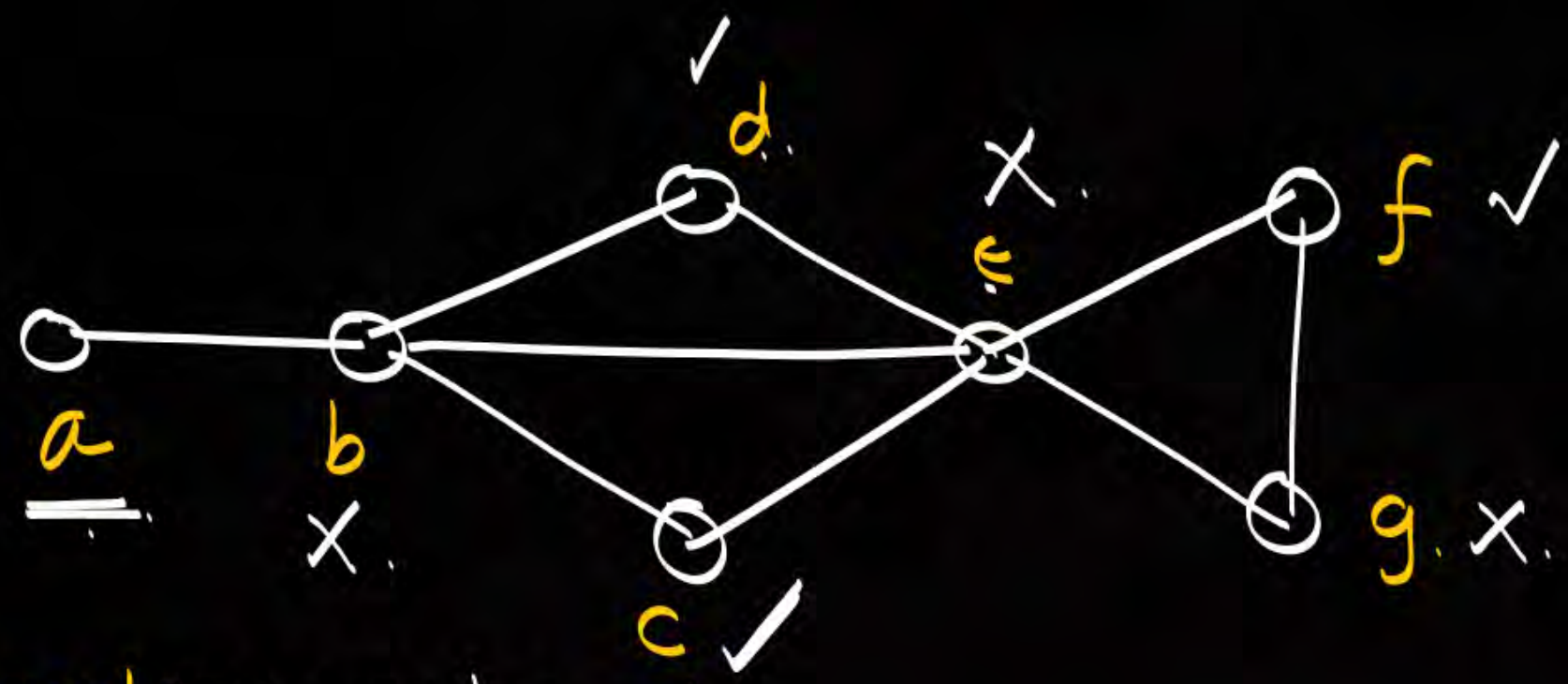


$$\chi(G) = 3.$$

$1 \rightarrow \{a, c, d, f\}$ set of non adjacent vertices
Independent set.

$2 \rightarrow \{b, g\}$

$3 \rightarrow \{e\}$



Independence set:
Set of non adjacent
vertices.

Independence set:

$\{a\}$ - I.S. ✓

$\{a, c\} \rightarrow$ I.S. ✓

$\{a, c, d\} \rightarrow$ I.S. ✓

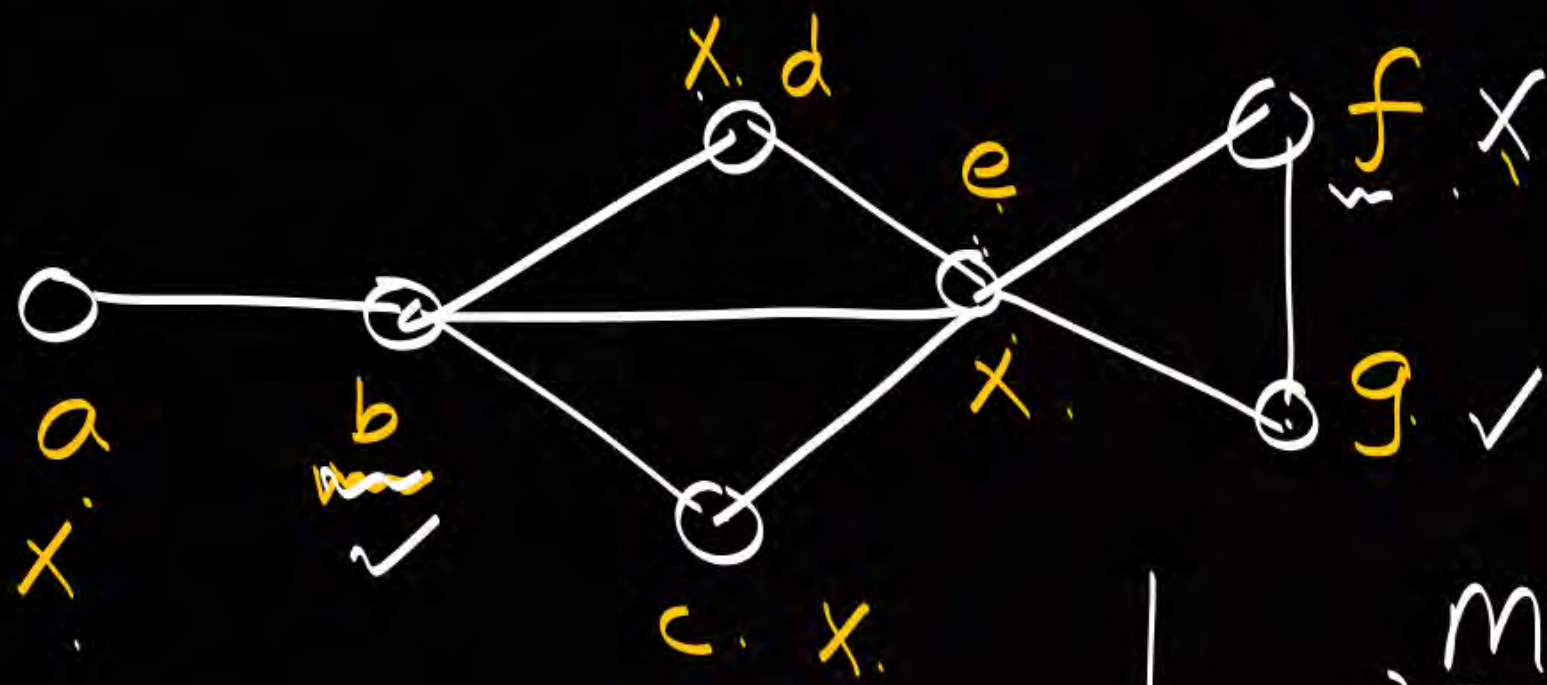
$\rightarrow \{a, c, d, f\} \rightarrow$ I.S.

$\{\}$ ✓

$\{a, b\}$ ✗

maximal
Independence
set

Independence set
such that we
cannot add
new vertex into this



$\{b\} \rightarrow \text{I.S.} \checkmark$

$\{b, g\} \rightarrow \text{I.S.}$
maximal Independence Set

MIS:

$\{a, c, d, f\}$

$\{a, c, d, g\}$

$\{b, f\}$

$\{b, g\}$

$\{a, e\}$

Independence no.

$\beta(G)$:

no. of vertices present in largest maximal Independence set

maximal
 not related to size but property such that we cannot add new vertex.

G_1



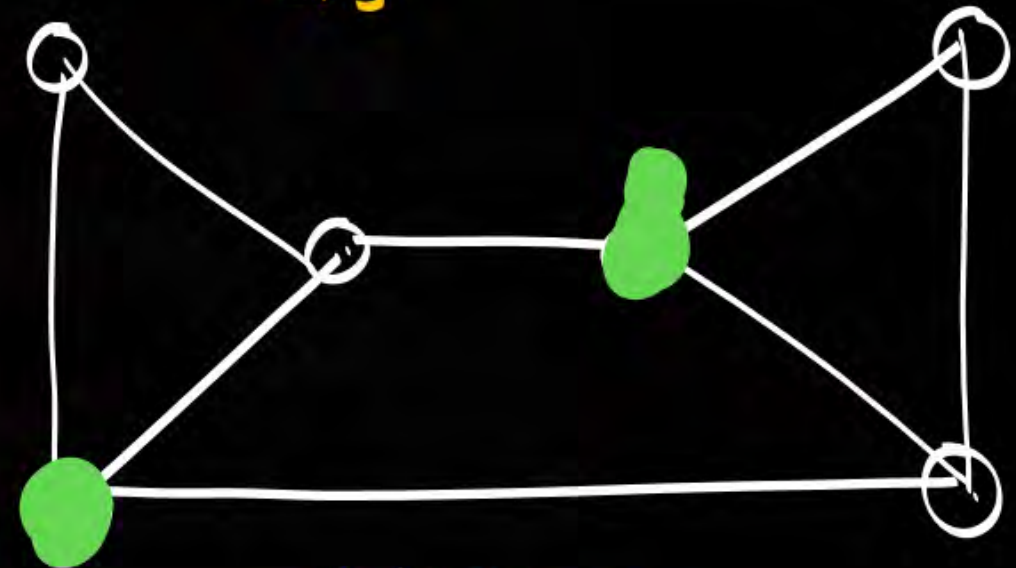
$$\beta(G) = 2.$$

G_2



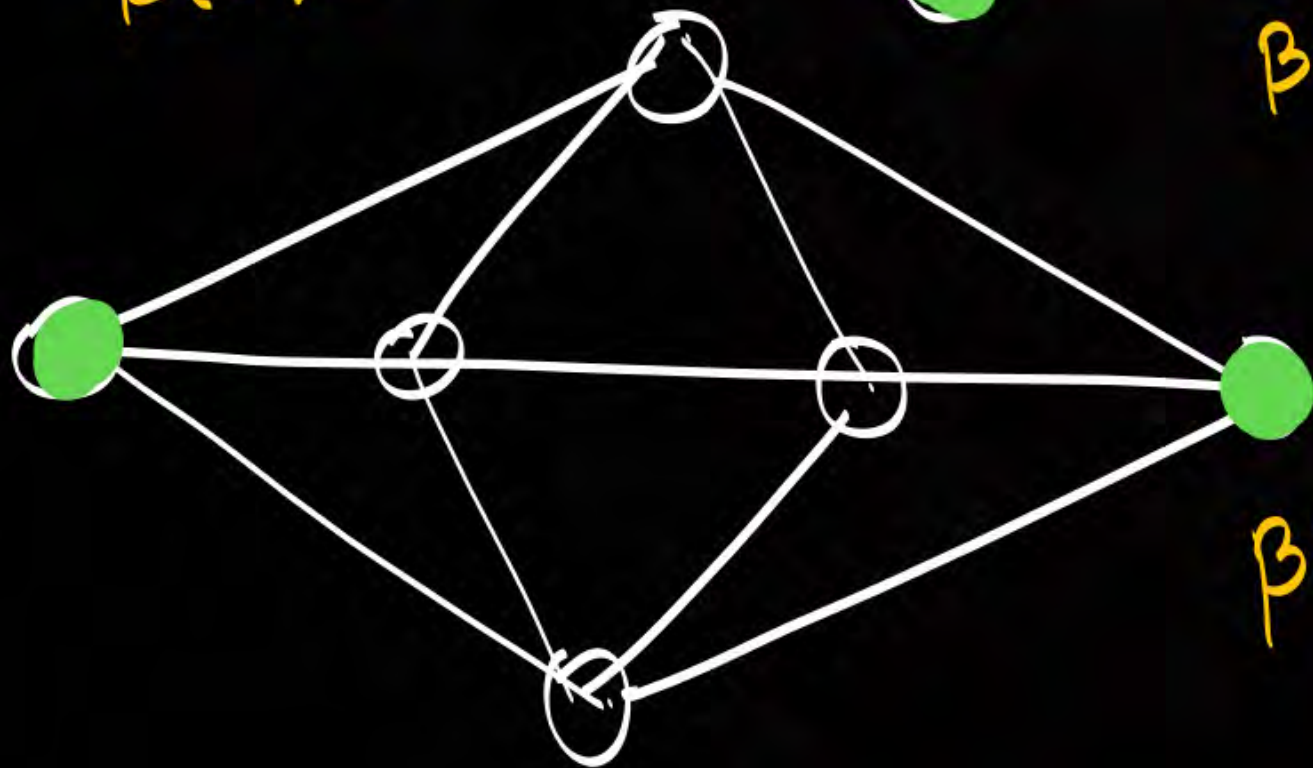
$$\beta(G) = 1.$$

G_3

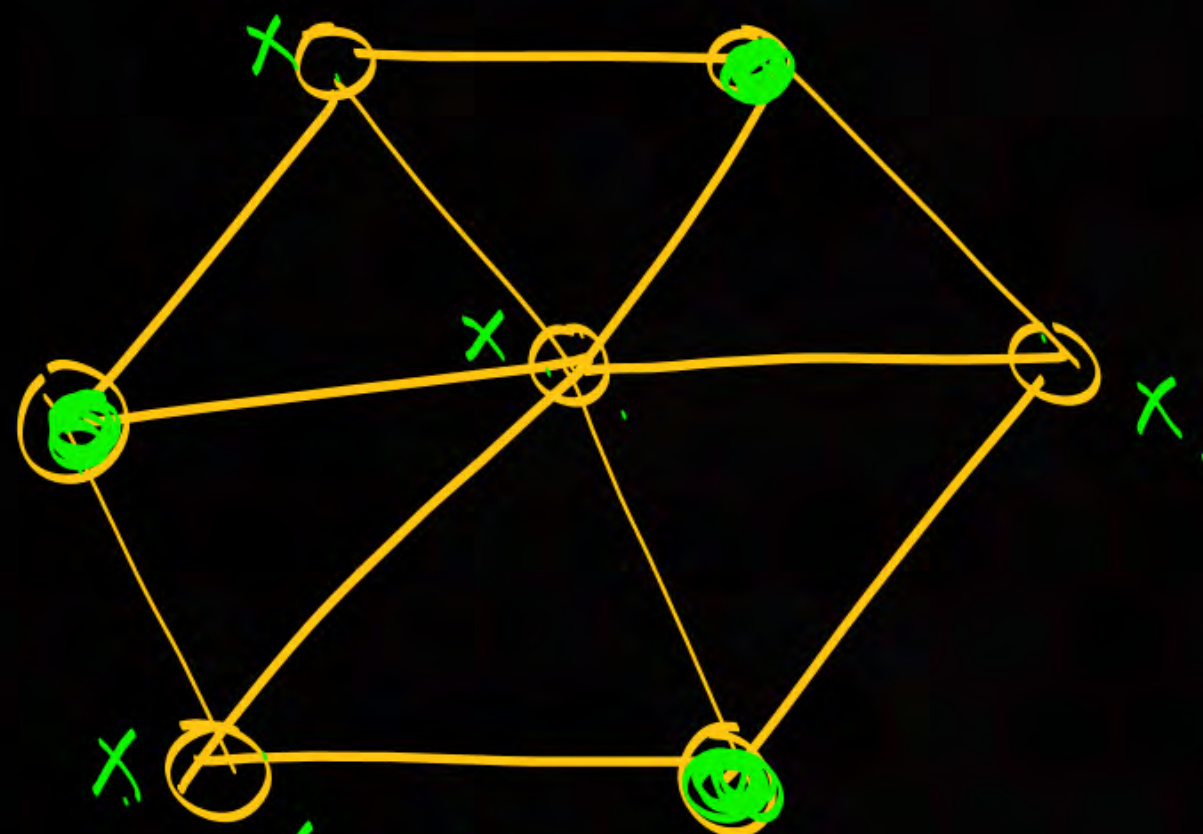


$$\beta(G) = 2$$

G_4



$$\beta(G) = 2$$

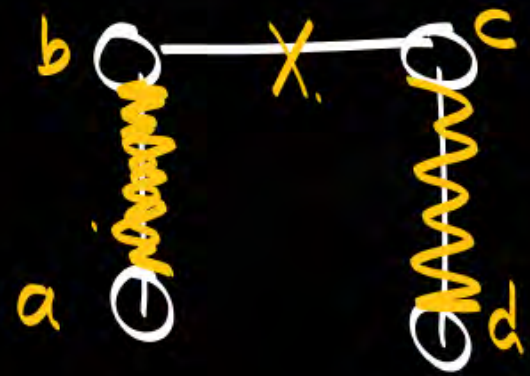


$$\beta(w_7) = 3.$$

(matching set).

Independence edge set:

Set of non adjacent edges.



$\{ab\}$ - ms.

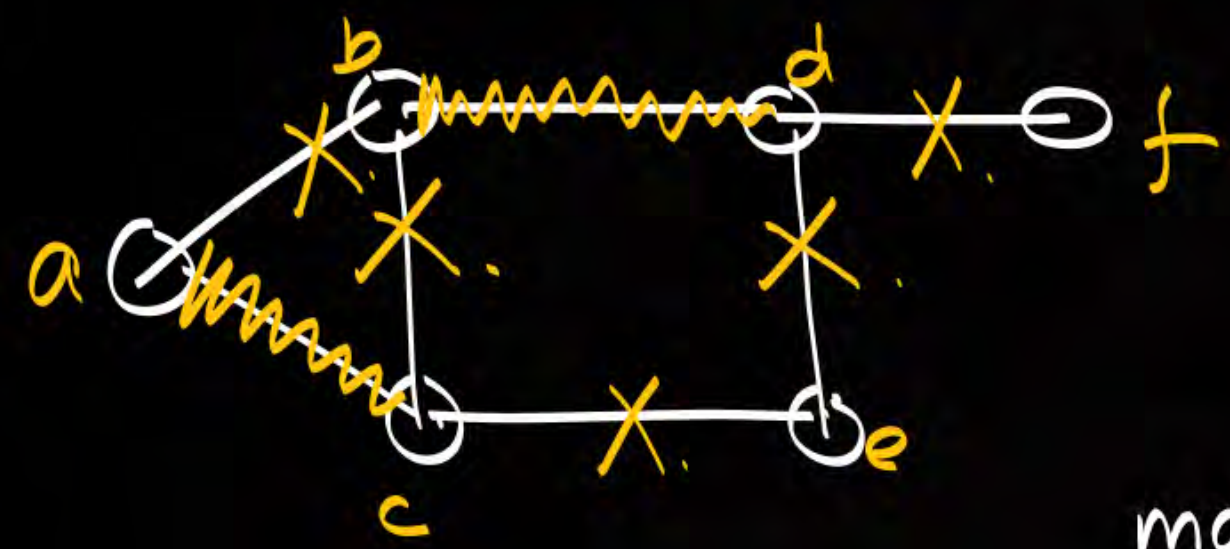
$\{ab, cd\} \rightarrow$

maximal matching set

maximal matching set

matching set

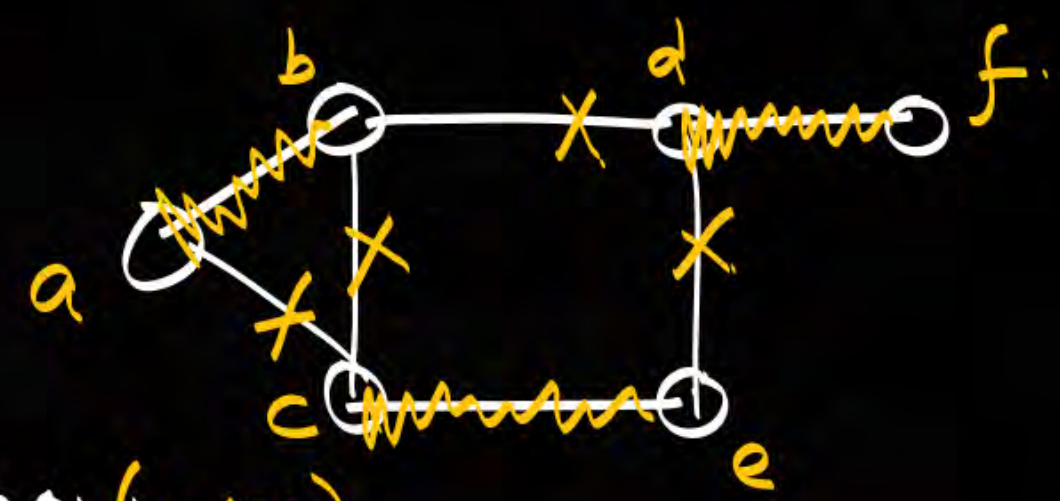
such that we cannot
add new edge into
this.



$\{bd\} \rightarrow ms$

$\{bd, ac\} \rightarrow$
maximal
matching
set.

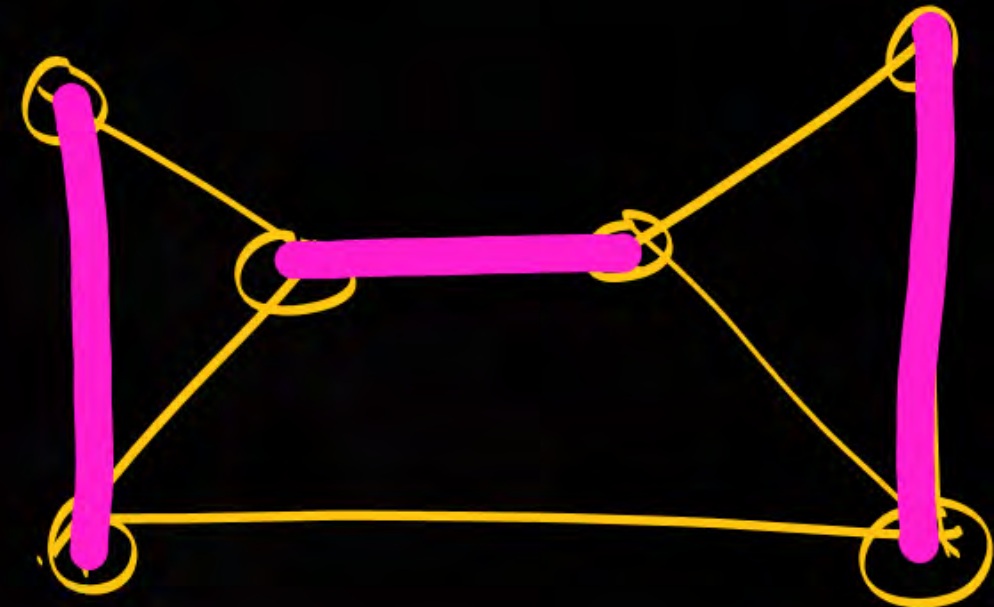
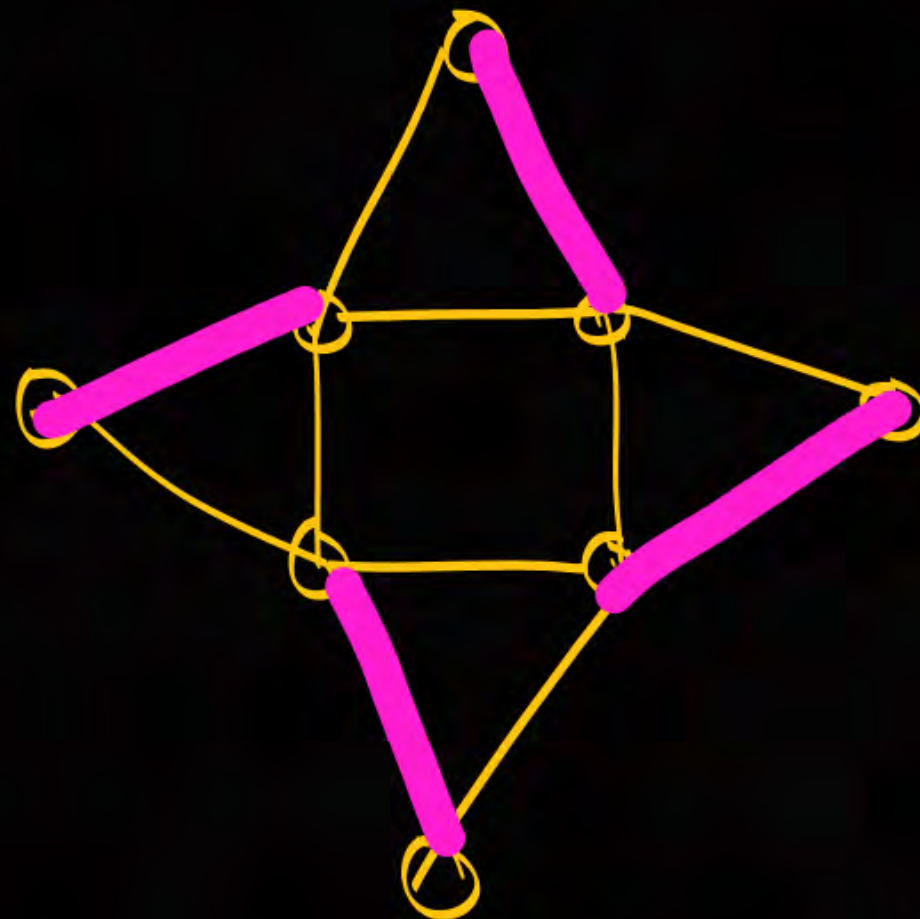
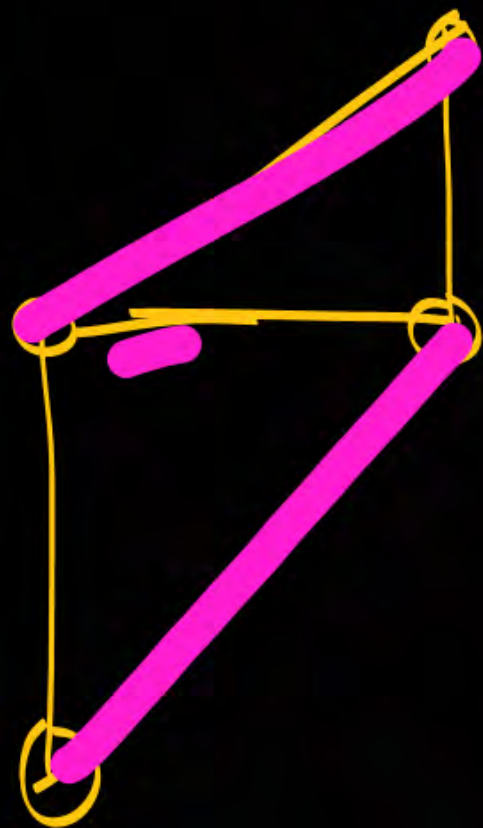
matching no. ($m(G)$)
no. of edges
present in
largest
maximal
matching
set



$\{df\} \rightarrow ms$

$\{df, ce\} \rightarrow ms$

$\{df, ce, ab\} \rightarrow$ maximal
matching
set.



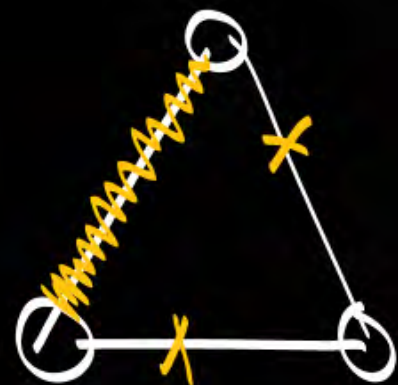
$$m(cn) = \left\lfloor \frac{c}{n} \right\rfloor$$

$$m(un) = \left\lfloor \frac{u}{n} \right\rfloor$$

$$m(kn) = \left\lfloor \frac{k}{n} \right\rfloor$$

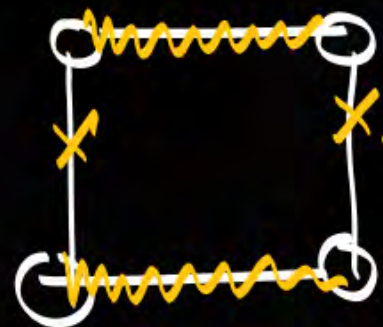
$$m(km, n) = \min(m, n)$$

C3



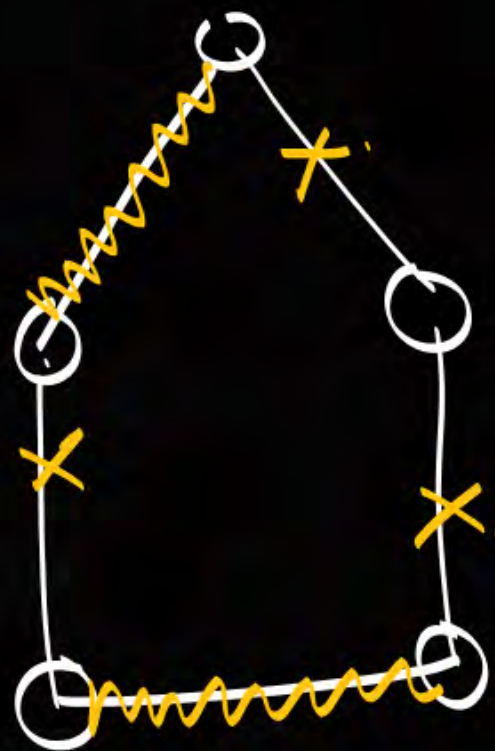
$$m(C_3) = 1.$$

C4



$$m(C_4) = 2.$$

C5

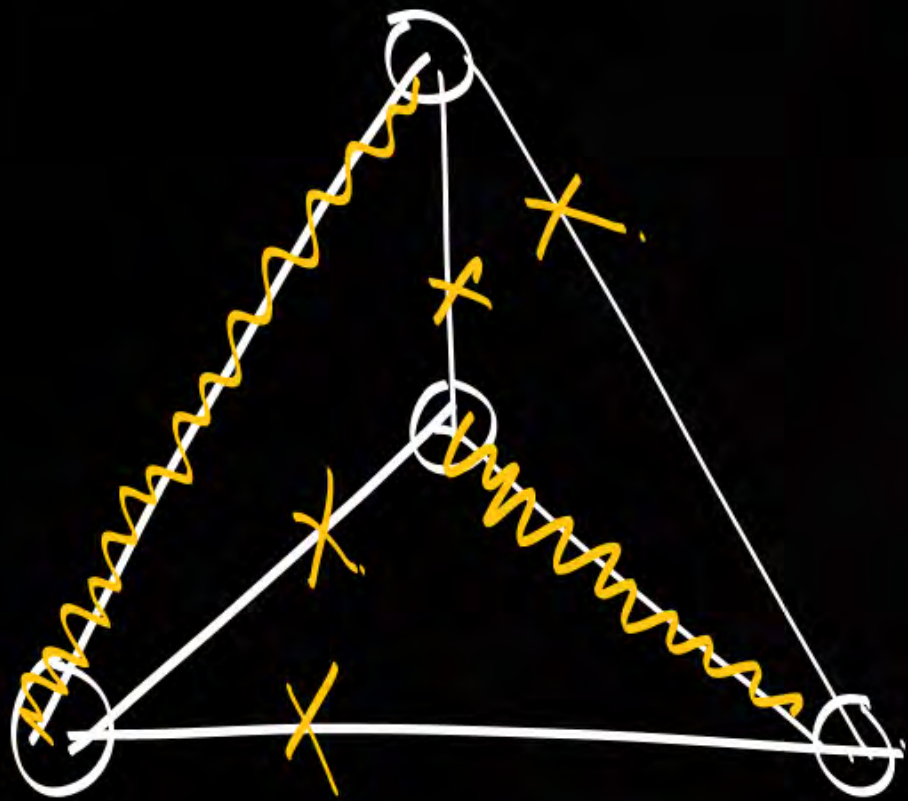


$$m(C_5) = 2$$

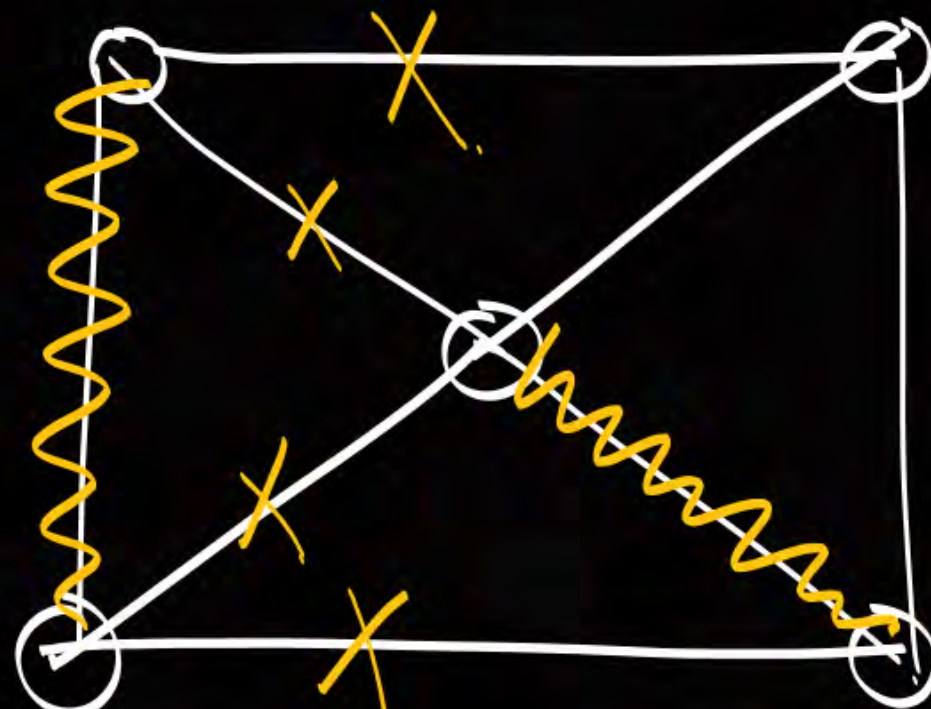
$$3/2 \rightarrow \lfloor 1.5 \rfloor = 1$$

$$\lfloor 4/2 \rfloor = 2.$$

$$\lfloor 5/2 \rfloor = 2.$$

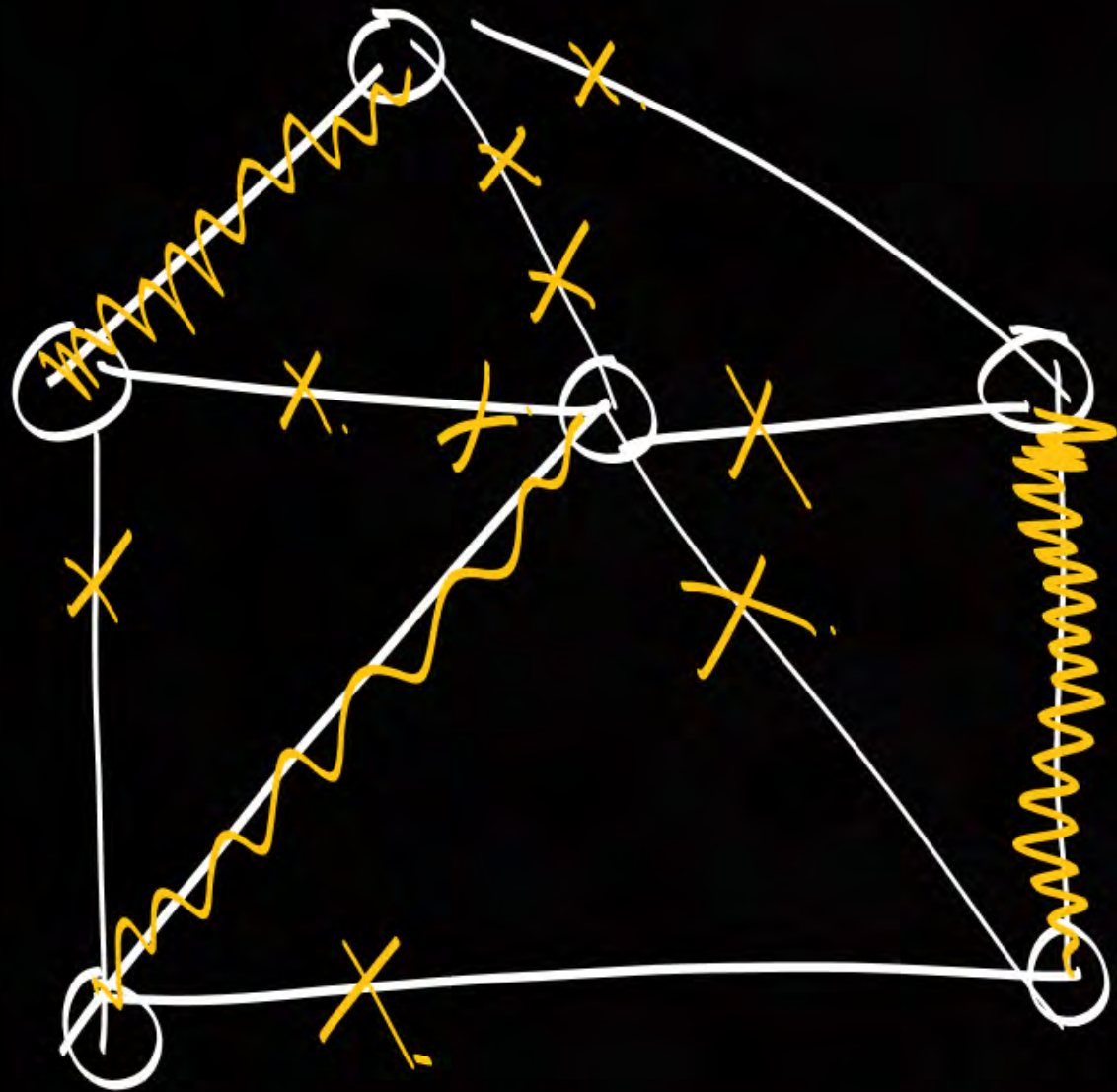


$$m(w_4) = 2.$$

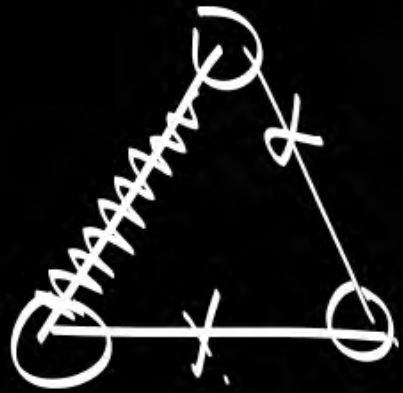


$$m(w_5) = 2.$$

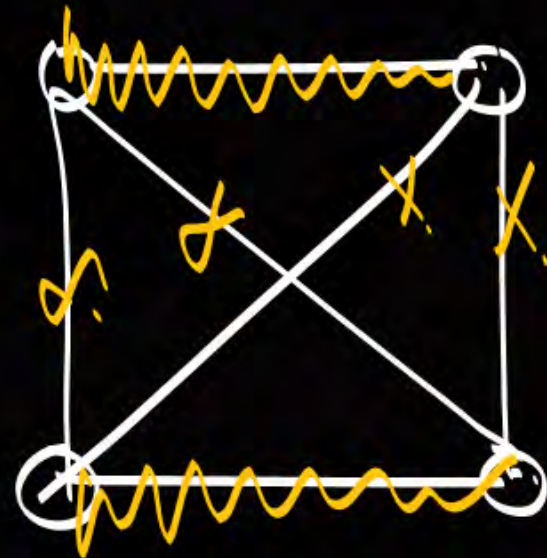
$$m(w_n) = \lfloor \frac{n}{2} \rfloor$$



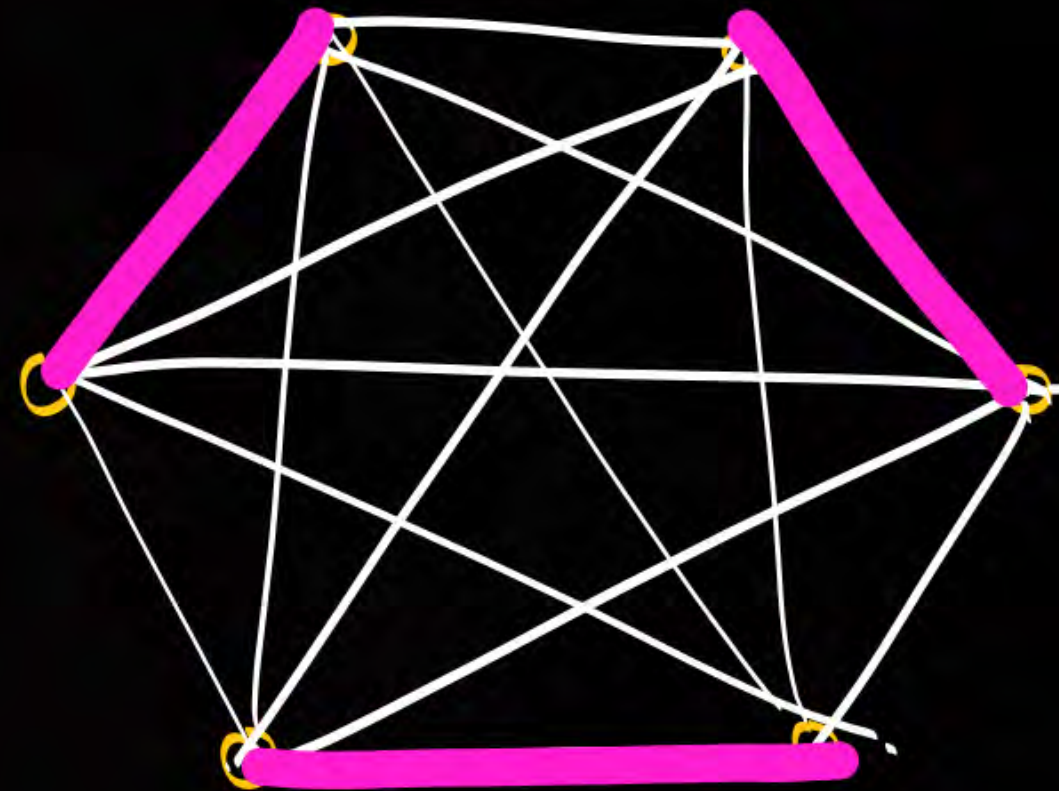
$$m(w_6) = 3.$$



$$m(K_3) = 1.$$



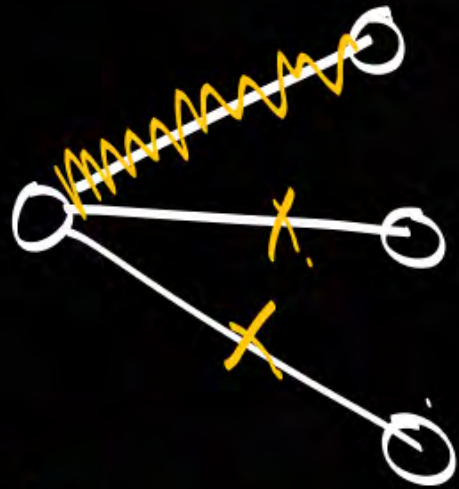
$$m(K_4) = 2$$



$$m(K_6) = 3.$$

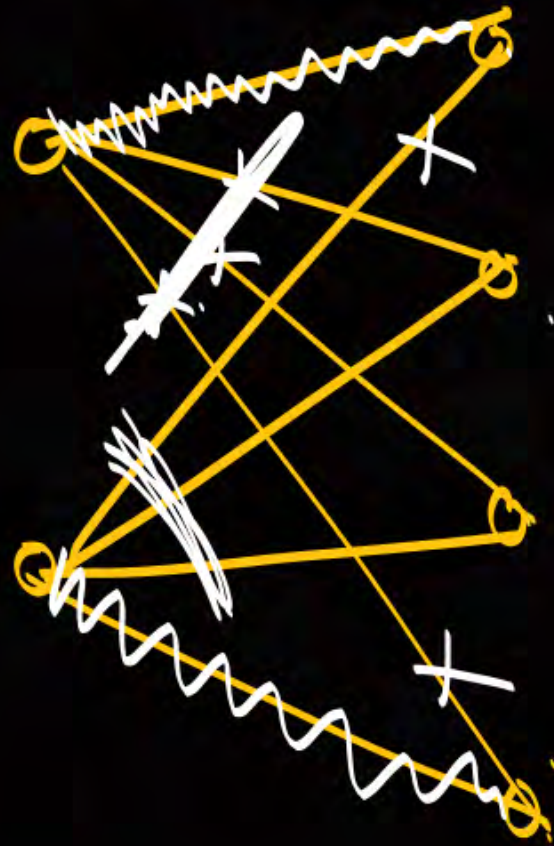
$$m(K_n) = \left\lfloor \frac{n}{2} \right\rfloor$$

$K_{1,3}$



$$m(K_{1,3}) = 1$$

$K_{2,4}$



$$m(2,4) = \min(2,4) = \underline{\underline{2}}$$



2 mins Summary

Topic

One

Dominating Set

Topic

Two

Matching Set

Topic

Three

Topic

Four

Topic

Five



THANK - YOU